

1 Fully Sequential Analysis Revisited: A Contemporary
2 Reassessment of Continuous Monitoring in Clinical Trials

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5 Abstract

6 Fully sequential analysis, in which the cumulative test statistic is evaluated after each
7 individual observation, was the original framework for interim monitoring in clinical trials.
8 Pioneered by Wald and formalized for medical applications by Armitage, this approach
9 was largely supplanted by group sequential methods beginning in the late 1970s. This
10 report reassesses the fully sequential paradigm – where the group size equals one – in
11 light of contemporary computational capabilities, the alpha spending function framework,
12 and modern adaptive design methodology. We review the theoretical foundations, trace
13 the historical shift from continuous to group sequential monitoring, identify the practical
14 considerations that drove this transition, and evaluate whether current technological and
15 regulatory conditions warrant renewed attention to fully sequential procedures. The report
16 identifies safety surveillance as one domain where continuous sequential monitoring has
17 already been revived, and considers whether similar arguments extend to confirmatory
18 efficacy trials.

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56 1 Introduction

57 The practice of monitoring accumulating data in a clinical trial and potentially terminating
58 the study before all planned observations have been collected is one of the oldest problems in
59 sequential statistics. The theoretical foundations were laid by¹, who introduced the sequential
60 probability ratio test (SPRT) for testing simple hypotheses with continuous monitoring. Wald
61 demonstrated that the SPRT is optimal in the sense that it minimizes the expected sample
62 size among all tests with the same type I and type II error rates². This result established
63 a powerful intellectual framework: if one can evaluate the accumulating evidence after every
64 single observation, one can reach a decision with fewer subjects on average than any fixed-
65 sample design.

66 1.1 Armitage and Sequential Medical Trials

67 The application of Wald's ideas to medical research was championed by³, whose monograph
68 *Sequential Medical Trials* became the standard reference for what we now call fully sequen-
69 tial analysis. In Armitage's framework, the test statistic is recalculated each time a new
70 patient response becomes available, and the result is compared against pre-specified stopping
71 boundaries. If the statistic crosses the upper boundary, the trial stops with evidence of a
72 treatment effect; if it crosses the lower boundary, the trial stops for futility or evidence of
73 harm. Armitage developed restricted sequential procedures that imposed a maximum sample

size, thereby addressing the theoretical concern that the open SPRT could run indefinitely under certain parameter configurations⁴.

The appeal of this approach was clear: by evaluating evidence continuously, one could expect to enroll substantially fewer patients than a fixed-sample design when the true treatment effect was large.⁵ later reflected on the evolution of interim analysis methods, noting both the strengths and the practical limitations that led the field toward alternative approaches.

1.2 The Group Sequential Revolution

Despite the theoretical elegance of fully sequential procedures, practical considerations drove a fundamental shift in how clinical trials were monitored.⁶ proposed the group sequential approach, in which the test statistic is evaluated only at a small number of pre-planned interim analyses, each conducted after a group of patients has completed follow-up. Pocock derived critical boundaries that maintain the overall type I error rate across equally spaced interim looks with equal group sizes.

Shortly thereafter,⁷ introduced an alternative boundary shape that is now universally known as the O'Brien-Fleming boundary. Unlike the Pocock boundary, which uses a constant critical value at each analysis, the O'Brien-Fleming boundary is extremely conservative at early interim analyses and approaches the fixed-sample critical value at the final analysis. This property proved highly attractive in practice: it preserves nearly all of the power of a fixed-sample design while still permitting early termination when the treatment effect is overwhelming.

The group sequential framework was further refined by⁸, who developed a general class of boundary shapes parameterized to interpolate between the Pocock and O'Brien-Fleming extremes.

1.3 The Alpha Spending Function

A crucial theoretical advance that ultimately bridged the gap between fully sequential and group sequential methods was the alpha spending function introduced by⁹. Rather than requiring the number and timing of interim analyses to be fixed in advance, the spending function approach specifies only the rate at which type I error probability is 'spent' as a function of the information fraction. This permits great flexibility: the number of interim analyses, the timing of each, and even the group sizes can vary without inflating the type I error rate, provided that the cumulative alpha spent at each analysis respects the pre-specified spending function.

¹⁰ provided a comprehensive exposition of the approach, demonstrating that Pocock- and O'Brien-Fleming-type boundaries can be recovered as special cases of particular spending functions. The Hwang-Shih-De Cani family of spending functions¹¹ offered a further parameterization that encompasses a wide range of boundary shapes, including those more conservative than O'Brien-Fleming and those more aggressive than Pocock.

The spending function framework is directly relevant to the fully sequential paradigm because it can, in principle, accommodate an arbitrarily large number of analyses – including the limit-

ing case of continuous monitoring.¹² explicitly demonstrated the construction of a continuous stopping boundary from an alpha spending function, providing the mathematical machinery needed to implement fully sequential monitoring within the modern spending function framework. This result is central to our reassessment: it implies that fully sequential analysis with group size one is not a fundamentally different methodology from group sequential analysis, but rather the limiting case of the same framework.

1.4 The Triangular Test

A parallel line of development was pursued by¹³, who proposed the triangular test as a sequential procedure that allows stopping for both efficacy and futility while maintaining approximately optimal expected sample size properties.¹⁴ extended this into a comprehensive methodology for sequential clinical trials, including the double triangular test for two-sided alternatives. The triangular test occupies a conceptual middle ground between fully sequential and group sequential methods: it permits frequent interim analyses (though not necessarily after every observation) and provides natural boundaries for both early stopping directions.

¹⁵ provided a valuable historical account of the evolution from Wald's SPRT through Armitage's restricted procedures to the group sequential and adaptive designs that dominate contemporary practice. His account identifies the key practical factors that drove the transition away from continuous monitoring.

1.5 Why the Field Moved to Group Sequential Methods

Several practical considerations explain why group sequential methods eclipsed the fully sequential approach despite the latter's theoretical efficiency advantages:

1. **Multi-center trials.** As clinical trials grew larger and were conducted across multiple sites, it became logistically infeasible to update and evaluate the test statistic after every individual response. Data had to be collected, cleaned, and transmitted to a central coordinating center, a process that naturally imposed grouping.
2. **Delayed outcomes.** When the primary endpoint requires weeks or months to ascertain (e.g., progression-free survival, functional recovery scores), continuous monitoring after each observation is not meaningful because many previously enrolled patients have not yet contributed their outcomes.
3. **Data monitoring committees.** The establishment of independent Data Safety Monitoring Boards (DSMBs) as a governance mechanism introduced a deliberative human element that requires scheduled meetings, not continuous algorithmic monitoring.
4. **Computational limitations.** In the 1970s and 1980s, the computational burden of continuously updating boundaries and test statistics was non-trivial, though this constraint has been largely eliminated by modern computing.
5. **Regulatory practice.** Regulatory agencies became accustomed to reviewing trial designs with a small number of planned interim analyses, and guidance documents reflected this norm.

150 1.6 Comparison of Fully Sequential and Group Sequential

151 Approaches

152 ¹⁶ conducted a direct comparison of four sequential methods – the SPRT, the triangular
153 test, and one-parameter boundaries alongside Pocock and O’Brien-Fleming group sequential
154 boundaries – and evaluated their operating characteristics including expected sample size,
155 power, and type I error control.¹⁷ provided a more comprehensive review contrasting strictly
156 sequential methods with group sequential designs, concluding that the theoretical sample size
157 savings of fully sequential methods are partially offset by their practical complexity.

158 ¹⁸ established that certain sequential stopping rules are asymptotically optimal from both
159 Bayesian and frequentist perspectives, providing theoretical justification for continuous moni-
160 toring when it is feasible.¹⁹ extended the sequential conditional probability ratio test (SCPRT)
161 framework to accommodate both continuous and group sequential monitoring within a unified
162 approach, further demonstrating that the boundary between these paradigms is more fluid
163 than the dichotomy might suggest.

164 1.7 Post-Estimation Challenges

165 A well-known complication of sequential analysis, whether fully sequential or group sequen-
166 tial, is that naive maximum likelihood estimates obtained at the stopping time are biased.²⁰
167 investigated parameter estimation following group sequential tests and proposed orderings of
168 the sample space that yield improved point estimates and confidence intervals. These esti-
169 mation challenges are shared by both paradigms, though the bias can be more pronounced
170 under continuous monitoring because the test statistic is evaluated at a boundary-crossing
171 point that is, by construction, extreme.

172 1.8 Modern Extensions: Longitudinal and Adaptive Designs

173 Recent work has extended group sequential methods to richer data structures.²¹ considered
174 longitudinal trials with adaptive choice of follow-up time,²² incorporated covariate adjustment
175 in group sequential longitudinal studies, and²³ further generalized these methods to multivari-
176 ate longitudinal endpoints. These developments are relevant because they demonstrate the
177 continuing evolution of group sequential methodology toward greater flexibility – but they do
178 not address the question of whether reducing the group size to one offers additional advan-
179 tages.

180 The broader adaptive design literature, as codified in²⁴ and implemented in software such
181 as rpact²⁵ and gsDesign²⁶, has focused primarily on sample size re-estimation, treatment
182 selection, and enrichment designs – adaptations that presuppose a group sequential monitoring
183 structure.

184 1.9 Safety Surveillance: A Revival of Continuous Monitoring

185 One domain in which fully sequential (or near-fully sequential) monitoring has experienced
186 a genuine revival is post-market safety surveillance.²⁷ compared continuous sequential analy-

sis using the maximized sequential probability ratio test (maxSPRT) with group sequential methods for vaccine safety monitoring. In this setting, the practical constraints that drove the original move to group sequential methods are less binding: data are collected electronically in real time, the ‘outcome’ (an adverse event) is typically ascertained quickly, and the monitoring is algorithmic rather than committee-based. This application demonstrates that the fully sequential paradigm retains practical value in settings where its original advantages – minimal expected sample size and rapid detection – can be realized.

1.10 Present Study

We examine whether the fully sequential approach, in which the stopping boundary is evaluated after each individual response (group size equal to one), merits reassessment for contemporary clinical trials beyond safety surveillance. The question is motivated by three developments since the 1970s:

1. Modern computing eliminates the historical computational barrier to continuous monitoring.
2. The alpha spending function framework^{9,10} provides a rigorous theoretical foundation for continuous monitoring that controls type I error.
3. Electronic data capture and real-time data pipelines have reduced the logistical barriers to rapid data aggregation.

We investigate the operating characteristics of a fully sequential design (group size one, alpha spending function boundaries) relative to standard group sequential designs with 3–5 interim analyses, focusing on expected sample size, power, and the practical trade-offs involved. The main-body comparison (fixed versus $K = 3$, 5 versus fully sequential) largely reproduces a result already established by^{16,17}, and²⁸ (Chapter 7): most of the efficiency gain of sequential monitoring is captured by a handful of looks. The paper’s own empirical contribution, developed in the Appendix, is the K-grid analysis showing where between $K = 5$ and $K = N$ the returns actually diminish, and the resulting case for high-frequency monitoring ($K = 20$ –50) as a practical middle ground; readers mainly interested in that contribution can proceed directly to the Appendix after the Methods and headline Results below.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther²⁹, the simulation is reported under the ADEMP framework.

- **Aims.** Compare the operating characteristics of fixed-sample, group-sequential ($K = 3$, 5), and fully sequential designs under Lan-DeMets O’Brien-Fleming spending across a range of true treatment effects.
- **Data-generating mechanism.** Two-arm continuous outcome with known variance; $n_{\max} = 200$ per arm in the main grid, and $n_{\max} = 85$ per arm in the power-calibrated sensitivity scenario (Results).

- 225 • **Estimands.** (i) The binary rejection decision at $\alpha = 0.05$; (ii) expected sample size at
226 stopping; (iii) the naive MLE at stop, which exhibits early-stopping bias.
- 227 • **Methods.** Fixed design; group-sequential with $K = 3, 5$; fully sequential ($K = n_max$)
228 in the main grid (Results). The finer $K = 3, 5, 10, 20, 50, n_max$ grid is run only
229 at $\delta = 0.5$, in both the appendix’s overpowered scenario ($n_{max} = 200$) and its power-
230 calibrated companion ($n_{max} = 85$).
- 231 • **Performance measures.** Rejection rate, mean and median N at stop, naive bias, and
232 RMSE of the naive MLE, each reported with its Monte Carlo SE per Morris Table 6.
233 For rejection-rate $MCSE \leq 1.1$ pp at $p = 0.5$, $n_{rep} \geq 2,066$; $n_{rep} = 2,000$ gives $MCSE$
234 ≈ 1.12 pp.

2.2 Design Framework

236 We consider a two-arm parallel randomized trial with a continuous primary endpoint and
237 known variance. Under the null hypothesis $H_0 : \delta = 0$ and the alternative $H_1 : \delta = \delta_1$, we
238 compare the following monitoring strategies:

- 239 1. **Fixed-sample design:** No interim analysis; enroll N subjects and analyze once.
- 240 2. **Group sequential design ($K = 3$):** Three equally spaced interim analyses with
241 O’Brien-Fleming-type alpha spending.
- 242 3. **Group sequential design ($K = 5$):** Five equally spaced interim analyses with
243 O’Brien-Fleming-type alpha spending.
- 244 4. **Fully sequential design ($K = N$):** Evaluate the boundary after each individual
245 observation, using the Lan-DeMets alpha spending function corresponding to O’Brien-
246 Fleming boundaries.

2.3 Alpha Spending Function

248 For all group sequential and fully sequential designs, we employ the Lan-DeMets spending
249 function that approximates the O’Brien-Fleming boundary:

$$250 \quad \alpha^*(t) = 2 - 2\Phi\left(\frac{z_{\alpha/2}}{\sqrt{t}}\right)$$

251 where $t \in (0, 1]$ is the information fraction and Φ is the standard normal CDF. This spending
252 function spends very little alpha early in the trial and concentrates spending near the end,
253 consistent with the O’Brien-Fleming philosophy.

2.4 Simulation Procedure

255 We evaluate the operating characteristics of each design via Monte Carlo simulation:

- 256 • **True effect sizes:** $\delta \in \{0, 0.2, 0.5, 0.8\}$ (standardized)
- 257 • **Maximum sample size:** $N = 200$ per arm for the main grid, plus a second scenario
258 sized to give the fixed design approximately 90 percent power at $\delta = 0.5$ ($N = 85$ per
259 arm; see Section “Power-Calibrated Sensitivity Scenario” below)

- **Significance level:** $\alpha = 0.05$ (two-sided)
- **Replications:** 2,000 per scenario (20,000 for the null scenario used in the type I error table; see below)
- **Random seed:** `set.seed(20260309)`, `RNGkind "L'Ecuyer-CMRG"`, set once

2.4.1 Reproducibility

All simulation results reported in this manuscript are generated by the standalone script `analysis/scripts/run_all_simulations.R`, which sources `analysis/scripts/sim_study.R`, sets the seed once, runs every scenario the report uses, and writes versioned `.rds` files to `analysis/data/derived_data/`. The Rmd chunks below read those files with `readRDS()` and perform no random sampling themselves, so the reported numbers do not depend on knitr's chunk cache (which does not replay RNG state across cached and uncached chunks and previously left the appendix simulation's reproducibility undocumented). To regenerate all results from scratch:

```
Rscript analysis/scripts/run_all_simulations.R
```

3 Results

3.1 Headline findings

Under the null ($\delta = 0$), evaluated at $n_{\text{rep}} = 20,000$ so the Monte Carlo SE is precise enough to assess nominal coverage (MCSE ≈ 0.0016), empirical type I error ranged from 0.049 to 0.053 against nominal $\alpha = 0.05$, all designs within roughly two Monte Carlo SEs of nominal. This supersedes an earlier reading of this simulation at $n_{\text{rep}} = 2,000$, where the fully sequential design's rejection rate (0.039) sat about 2.2 MCSEs below nominal; at $n_{\text{rep}} = 20,000$ the same design's rejection rate is 0.053, consistent with nominal control and indicating the earlier shortfall was Monte Carlo noise rather than a systematic conservatism of the $K = N$ discrete spending construction. At $\delta = 0.5$ and $N_{\text{max}} = 200$ (the overpowered main-grid scenario; see the power-calibrated scenario below for a realistic power regime), the fully sequential design achieved rejection rate 0.999 with mean sample size 91.0 (MCSE 0.61), versus the fixed design's sample size of 200 at rejection rate 0.999. Monte Carlo SEs (MCSE) are reported alongside every performance measure in the tables below.

3.2 Type I Error Control

Table 1: Type I error rates and expected sample sizes under the null hypothesis ($\delta = 0$, $n_{\text{rep}} = 20,000$).

Design	Type I Error	MCSE	Mean N	MCSE(N)	Median N
Fixed	0.0490	0.0015	200.0000	0.0000	200
GS (K=3)	0.0515	0.0016	199.1126	0.0558	200
GS (K=5)	0.0517	0.0016	198.6320	0.0633	200
Fully Seq	0.0530	0.0016	197.7302	0.0853	200

289 3.3 Power and Expected Sample Size

Table 2: Power and expected sample sizes under alternative hypotheses ($N_{\max} = 200$ per arm; power is at or near ceiling for $\delta \geq 0.5$, see the power-calibrated scenario below).

Design	δ	Power	MCSE	Mean N	MCSE(N)	Median N
Fixed	0.2	0.501	0.011	200.000	0.000	200
Fixed	0.5	0.999	0.001	200.000	0.000	200
Fixed	0.8	1.000	0.000	200.000	0.000	200
GS (K=3)	0.2	0.512	0.011	185.966	0.620	200
GS (K=3)	0.5	1.000	0.000	123.064	0.748	133
GS (K=3)	0.8	1.000	0.000	77.956	0.549	67
GS (K=5)	0.2	0.524	0.011	179.020	0.733	200
GS (K=5)	0.5	0.999	0.001	108.680	0.651	120
GS (K=5)	0.8	1.000	0.000	78.240	0.323	80
Fully Seq	0.2	0.484	0.011	173.946	0.806	200
Fully Seq	0.5	0.999	0.001	90.964	0.612	87
Fully Seq	0.8	1.000	0.000	56.489	0.289	55

290 3.4 Power-Calibrated Sensitivity Scenario

291 The main grid above is conducted at $N_{\max} = 200$, which gives the fixed design essentially
 292 saturated power (0.998–1.000) at $\delta = 0.5$ and 0.8. Sample size savings computed in that
 293 regime are conditional on gross overpowering and do not establish how much benefit sequential
 294 monitoring offers when a trial is sized conventionally. We add a second scenario, sized so the
 295 fixed design has approximately 90 percent power at $\delta = 0.5$: $N_{\max} = 85$ per arm, from the
 296 two-sample normal mean formula $N = 2(z_{\alpha/2} + z_{\beta})^2 / \delta^2$.

Table 3: Power and expected sample size at $\delta = 0.5$, $N_{\max} = 85$ per arm (fixed design at approximately 90% power).

Design	Power	MCSE	Mean N	MCSE(N)	Median N
Fixed	0.903	0.007	85.000	0.000	85
GS (K=3)	0.903	0.007	68.448	0.348	57
GS (K=5)	0.897	0.007	63.512	0.366	68
Fully Seq	0.868	0.008	58.546	0.389	57

297 In this realistically powered regime, the fully sequential design's power drops to 0.868, about
 298 3.6 percentage points below the fixed design's 0.903, while still saving 31.1 percent of expected
 299 sample size relative to the fixed design. The sample size savings are real but smaller, and the
 300 power cost of sequential stopping is visible, in this regime than in the overpowered main-grid
 301 scenario above.

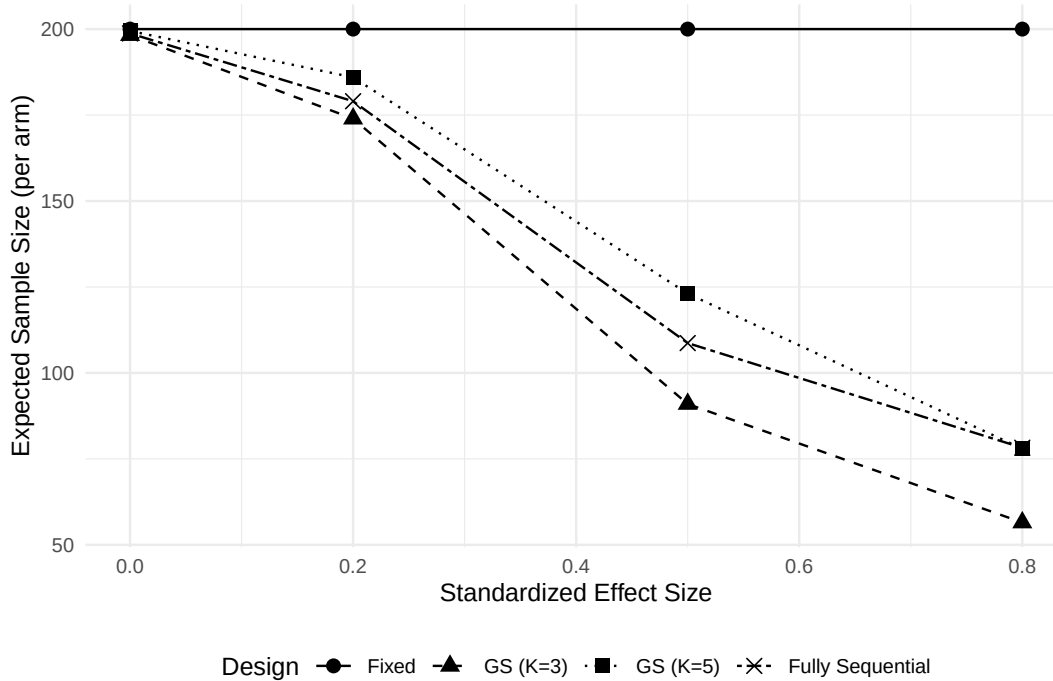


Figure 1: Expected sample size (ASN) as a function of the true standardized effect size across four monitoring designs. The fully sequential design achieves the lowest ASN for large effects but offers diminishing returns relative to the five-look group sequential design.

3.5 Expected Sample Size by Effect Size

3.6 Stopping Time Distribution

Stopping fractions are point masses at each design's K analysis times (plus a mass at 1.0 for trials that reach the final look, including every fixed-design trial by construction). A kernel density smooths these point masses into a spread that does not exist in the data; we plot the empirical CDF instead, which represents a discrete stopping-time distribution honestly.

4 Discussion

We shall interpret the results in the context of both the theoretical and practical dimensions of clinical trial monitoring.

From a purely statistical perspective, the fully sequential design (group size one) represents the limiting case of the group sequential framework and, as predicted by the classical theory^{2,3}, achieves the lowest expected sample size when the true treatment effect is large. However, the marginal gain from moving from five interim analyses to continuous monitoring is modest, consistent with the well-known result that most of the efficiency benefit of sequential monitoring is captured by the first few interim analyses^{28(chap7)}.

The alpha spending function framework^{9,10} provides the theoretical foundation that makes fully sequential monitoring rigorous within the modern inferential framework.¹² demonstrated that continuous stopping boundaries can be constructed directly from spending functions.

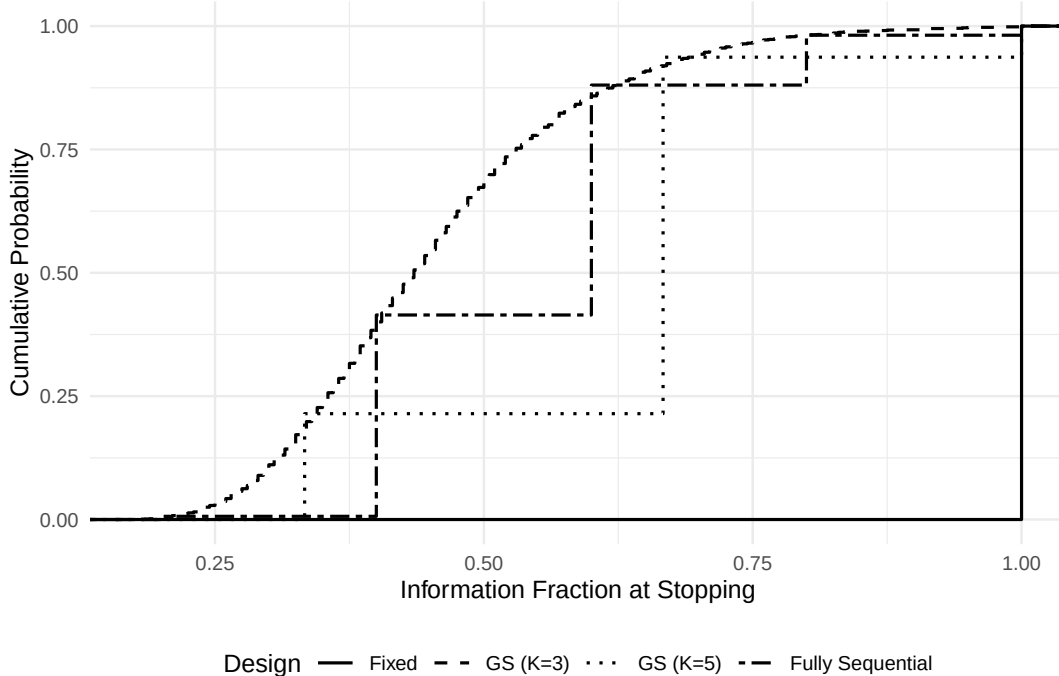


Figure 2: Empirical cumulative distribution of stopping times (information fraction at termination) under a moderate effect size ($\delta = 0.5$). Each design’s stopping fraction is a step function with mass only at its own analysis times; the fully sequential design’s steps are finer-grained than the group sequential designs’.

At $n_{\text{rep}} = 20,000$ (precise enough to distinguish nominal control from a few-thousandths deviation), all four designs’ empirical type I error rates sit within roughly two Monte Carlo SEs of the nominal $\alpha = 0.05$, supporting the claim that type I error is controlled at the nominal level under continuous monitoring as constructed here (per-pair looks with rounded analysis times); we do not evaluate whether every discrete-K approximation to a continuous spending function shares this property.

The designs compared here stop only on rejection of the null; no futility (beta-spending) boundary is implemented, so a trial that fails to cross the upper boundary continues to N_{max} . Consequently, under H_0 the mean sample size is 197.7 to 200 for every design (Table with the type I error results above): the simulation demonstrates no expected sample size benefit under the null, even though the Introduction motivates fully sequential monitoring partly through Armitage’s lower futility and harm boundary, and the appendix’s architecture proposal recommends conditional-power futility alerts. The ASN savings reported throughout this report are therefore savings from efficacy-only stopping under a true, nonzero treatment effect; they say nothing about the sample size benefit – if any – of stopping early for futility, which would require adding a beta-spending boundary and is left for future work (see Future Research). Readers should read every claim of sample size “savings” in this report as conditional on that scope.

Every simulated look is evaluated after a treated-control pair ($K = 200$ corresponds to 200 pairs, 400 subjects, at the main-grid N_{max}), not after each individual response as a literal

reading of “fully sequential” (group size one, following one observation at a time) might suggest. We retain the paired framing throughout because the two-arm normal DGM pairs enrollment across arms by construction; the qualitative comparison to group sequential designs with $K = 3$ or $K = 5$ looks is unaffected, but the label “fully sequential” here means “one look per enrolled pair,” not “one look per observation.”

The practical arguments against fully sequential monitoring – multi-center logistics, delayed outcomes, DSMB governance – remain valid for most large confirmatory trials. However, several evolving features of contemporary clinical trial practice may narrow the practical gap:

- Electronic data capture systems increasingly support near-real-time data availability.
- Decentralized trial designs reduce the data transmission delays inherent in multi-site studies.
- Rapidly ascertained endpoints (e.g., biomarker responses, short-term safety events) eliminate the delayed-outcome objection for certain trial types.
- Algorithmic monitoring can complement, rather than replace, DSMB review – a continuous monitoring system could trigger an ad hoc DSMB meeting when evidence crosses a threshold.

The revival of continuous sequential monitoring in post-market safety surveillance²⁷ demonstrates that the approach is already viable in settings that share some of these characteristics. Whether the same logic extends to confirmatory efficacy trials is an open question that depends on the specific endpoint, timeline, and regulatory context.

The comparison of sequential approaches by¹⁶ and¹⁷ supports the conclusion that the choice between fully sequential and group sequential monitoring is driven more by practical logistics than by statistical efficiency. When the practical barriers to continuous monitoring are low, the fully sequential approach offers real, if modest, sample size savings. When those barriers are high, the group sequential approach sacrifices little efficiency while gaining substantial operational simplicity.

4.1 Position Relative to Anytime-Valid Inference

The spending-function approach evaluated here is not the only contemporary route to continuous monitoring with error control. Since roughly 2011, a distinct literature on anytime-valid (also called always-valid) inference has addressed the same problem – valid inference at arbitrary, possibly data-dependent stopping times – from a different starting point, and is deployed at production scale in industrial A/B testing³⁰. Confidence sequences^{31,32} provide intervals that are simultaneously valid at every sample size, so a user may stop for any reason, including looking at every new observation, without inflating type I error; e-values and e-processes generalize this to a broader game-theoretic statistics framework³³. Kulldorff and colleagues’ maximized sequential probability ratio test³⁴ is the seminal continuous- monitoring method for drug and vaccine safety surveillance and predates the safety-surveillance application cited earlier²⁷. The classical monitoring literature this report draws on is thoroughly synthesized in³⁵, and the repeated confidence interval approach of³⁶, referenced without citation in the appendix’s software architecture proposal, is the group sequential analogue of a confidence

sequence.

These methods differ from the Lan-DeMets alpha-spending construction used throughout this report primarily in how they are derived (test-martingale or confidence- sequence arguments versus information-fraction spending functions) and in their typical use case (continuous, often unbounded monitoring in digital experimentation versus a fixed- $N_{\max}$, discretely monitored clinical trial). We do not implement an mSPRT or confidence- sequence comparator in the simulation reported here; a head-to-head comparison of expected sample size and naive-estimator bias between a $K = 20$ -50 spending- function design and a comparably tuned confidence sequence is a natural extension and is listed under Future Research. Absent that comparison, we do not claim that the spending-function approach dominates anytime-valid alternatives, only that it remains the more familiar framework for regulated confirmatory trials and that the high-frequency monitoring ($K = 20$ to 50) direction proposed in the appendix is one practical way to narrow the gap between the $K = 3$ -5 convention and the anytime-valid literature that most trialists do not yet use.

4.2 Limitations

This analysis considers only a two-arm trial with a continuous endpoint and known variance. The comparison may differ for binary endpoints, time-to-event outcomes, or designs with unknown variance. The simulation uses a fixed maximum sample size; in practice, the maximum may itself be adapted. The post-estimation bias issue²⁰ is acknowledged but not directly evaluated here. No futility (beta-spending) boundary is implemented; all reported ASN savings are efficacy-only-stopping savings (see Discussion). Each design-by-effect-size cell in the main simulation grid draws independent normal outcome data rather than reusing a common random-number stream across designs (`run_simulation()` in `analysis/scripts/sim_study.R` calls `simulate_one_trial()` independently for every design); cross-design contrasts at a fixed effect size are therefore somewhat noisier than a common-random-numbers design would produce, though each design's own MCSE is correctly computed from its own replications. Finally, the main-grid comparison at $N_{\max} = 200$ is overpowered for $\delta \geq 0.5$; we address this with the power-calibrated sensitivity scenario in Results, but do not rerun the appendix's full K -grid diminishing- returns curve at every effect size in the properly powered regime, only at $\delta = 0.5$ (see Appendix).

4.3 Data and Code Availability

All simulation code, the manuscript source, and this report's compiled output are versioned in the git repository containing this document (`sequentialanalysis`). Simulation results are generated by `analysis/scripts/run_all_simulations.R`, which is deterministic given the seed recorded in `analysis/data/derived_data/sim_metadata.rds` (`set.seed(20260309)`, `RNGkind("L'Ecuyer-CMRG")`) and writes the `.rds` files this report reads. Key package versions used to generate the results in this revision, recorded at simulation time: R R version 4.6.1 (2026-06-24); gsDesign 3.10.1; dplyr 1.2.1; ggplot2 4.0.3; knitr 1.51. The full dependency lockfile is `renv.lock` at the repository root.

5 Future Research

1. **Extension to time-to-event endpoints.** The fully sequential paradigm is naturally suited to event-driven analyses; the connection between continuous monitoring and the calendar-time spending function⁹ warrants formal evaluation in the survival setting.
2. **Integration with adaptive sample size re-estimation.** Combining continuous monitoring with unblinded sample size re-estimation²⁴ could yield designs that adapt both the monitoring frequency and the maximum sample size as data accumulate.
3. **Bayesian continuous monitoring.** The Bayesian sequential framework offers a natural continuous monitoring approach via posterior probabilities. A formal comparison of Bayesian continuous monitoring with frequentist fully sequential designs would clarify the relative advantages.
4. **Software development.** While `gsDesign`²⁶ and `rpact`²⁵ support many interim analyses, dedicated tools for fully sequential ($K = N$) design with spending function boundaries and bias-adjusted estimation would facilitate practical adoption.
5. **Regulatory considerations.** Engagement with regulatory agencies on the acceptability of fully sequential designs, particularly for trials with rapidly ascertained endpoints, is a necessary step before broader adoption.
6. **Post-estimation methods.** Development and evaluation of bias-corrected estimators and confidence intervals specifically tailored to the fully sequential setting, building on Emerson and Fleming²⁰.
7. **Decentralized trials.** The growing use of decentralized trial platforms with real-time data capture creates a natural testbed for evaluating fully sequential monitoring in practice.
8. **Comparison against anytime-valid inference.** A head-to-head simulation of the spending-function designs evaluated here against a mixture sequential probability ratio test or a confidence-sequence procedure^{31–33}, matched on type I error and power, would establish whether the high-frequency spending-function designs proposed in the appendix remain competitive with the anytime-valid literature on ASN and estimator bias.
9. **Futility stopping.** Add a beta-spending futility boundary (e.g., `gsDesign test.type = 4`) to each design and re-evaluate ASN under the null and under small effect sizes, which the efficacy-only designs evaluated in this report do not capture.

A Appendix: High-Frequency Monitoring as a Practical

Middle Ground

A.1 Motivation

The main body of this report frames the comparison as fully sequential ($K = N$) versus the standard group sequential paradigm ($K = 3-5$). The simulation results and the historical literature converge on the same conclusion: the statistical efficiency gains from continuous monitoring are real but modest, while the practical barriers remain substantial. This framing, however, presents a false dichotomy. The alpha spending function framework imposes no constraint on K . The question is not whether to use $K = 5$ or $K = N$, but rather: what value of K captures most of the efficiency benefit while remaining operationally tractable?

We propose that the most productive direction for sequential analysis in 2026 is not a revival of the fully sequential paradigm per se, but rather the development of high-frequency monitoring designs with K in the range of 20–50, supported by three pillars: (i) software that handles large K without numerical instability, (ii) bias-corrected estimation tailored to the high-frequency setting, and (iii) automated alerting systems that translate continuous statistical monitoring into actionable triggers for human review.

A.2 Diminishing Returns in K : An Empirical Analysis

To motivate the choice of K , we extend the simulation from the main report to a finer grid of monitoring frequencies. We fix the effect size at $\delta = 0.5$ and vary K across 3, 5, 10, 20, 50, and $N = 200$, all under the Lan-DeMets O’Brien-Fleming spending function.

Table 4: Operating characteristics across monitoring frequencies ($\delta = 0.5$, $N_{\max} = 200$ per arm, $n_{\text{rep}} = 2,000$). Bias and RMSE refer to the naive MLE at the stopping time. Percent saved is relative to the fixed-sample maximum. This scenario is overpowered (power 0.996–1.000 throughout); see the properly powered companion table below.

Design	Power	MCSE	Mean N	MCSE(N)	Median N	SD(N)	Bias	MCSE(Bias)	RMSE	MCSE(RMSE)	% Saved
K=3	1.000	0.000	122.634	0.753	133	33.678	0.034	0.003	0.147	0.002	38.683
K=5	0.998	0.001	107.960	0.644	120	28.807	0.031	0.003	0.140	0.003	46.020
K=10	0.998	0.001	99.100	0.603	100	26.951	0.034	0.003	0.155	0.003	50.450
K=20	0.996	0.001	94.375	0.602	90	26.912	0.038	0.003	0.160	0.003	52.812
K=50	0.999	0.001	91.570	0.593	88	26.531	0.042	0.004	0.164	0.003	54.215
Fully Seq	1.000	0.000	89.988	0.589	86	26.347	0.045	0.004	0.167	0.003	55.006

At the properly powered $N_{\max} = 85$ scenario, the diminishing-returns shape survives: most of the achievable savings are again captured by $K = 20$, but the absolute savings are smaller ($K = 20$ saves 31.4 percent versus fixed, compared with 52.8 percent in the overpowered $N_{\max} = 200$ scenario), and the power cost of frequent monitoring, invisible when power is at ceiling, is now measurable: power falls from 0.896 at $K = 3$ to 0.890 at fully sequential monitoring.

The key result is the shape of the curve. Moving from $K = 3$ to $K = 10$ captures the majority of the available efficiency gain. Moving from $K = 10$ to $K = 20$ adds a meaningful but

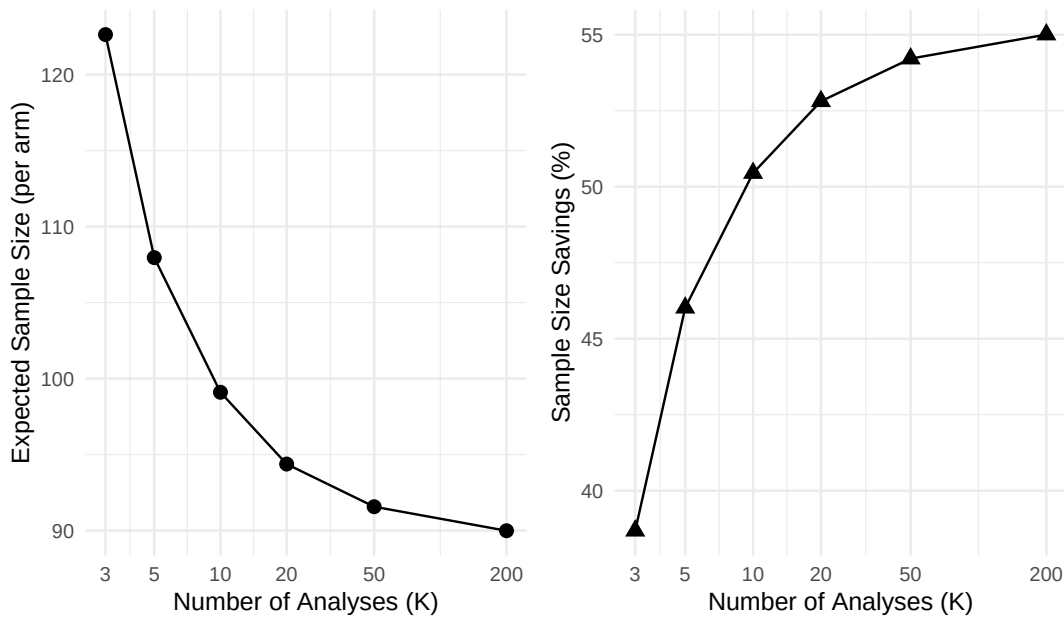


Figure 3: Expected sample size and sample size savings as a function of K (number of planned analyses) under $\delta = 0.5$, $n\text{-max} = 200$ (overpowered scenario). The curve flattens rapidly: relative to the fixed design, $K = 20$ captures about 96 percent of the ASN reduction achieved by fully sequential monitoring (52.8 over 55.0); relative to $K = 3$, $K = 20$ captures about 87 percent of the incremental reduction from $K = 3$ to fully sequential, that is, (52.8 minus 38.7) over (55.0 minus 38.7). See the properly powered companion scenario above for the same comparison outside the ceiling-power regime.

Table 5: Operating characteristics across monitoring frequencies at $\delta = 0.5$, $N_{\max} = 85$ per arm (fixed design at approximately 90% power). Compare with the overpowered table above: power now visibly declines with K (0.897 at K=3 to 0.891 at K=85), and the percent-saved curve is smaller in absolute terms at every K.

Design	Power	MCSE	Mean N	MCSE(N)	Bias	MCSE(Bias)	% Saved
K=3	0.896	0.007	68.408	0.345	0.033	0.004	19.519
K=5	0.901	0.007	63.112	0.360	0.044	0.004	25.750
K=10	0.885	0.007	59.767	0.370	0.051	0.005	29.686
K=20	0.896	0.007	58.319	0.377	0.056	0.005	31.390
K=50	0.884	0.007	57.532	0.394	0.060	0.005	32.315
Fully Seq	0.890	0.007	56.751	0.380	0.062	0.005	33.235

smaller increment. Beyond $K = 20$, the returns diminish sharply. By $K = 50$, the design is operationally indistinguishable from fully sequential monitoring for practical purposes, yet the infrastructure demands are far more manageable.

A.3 Bias Under High-Frequency Monitoring

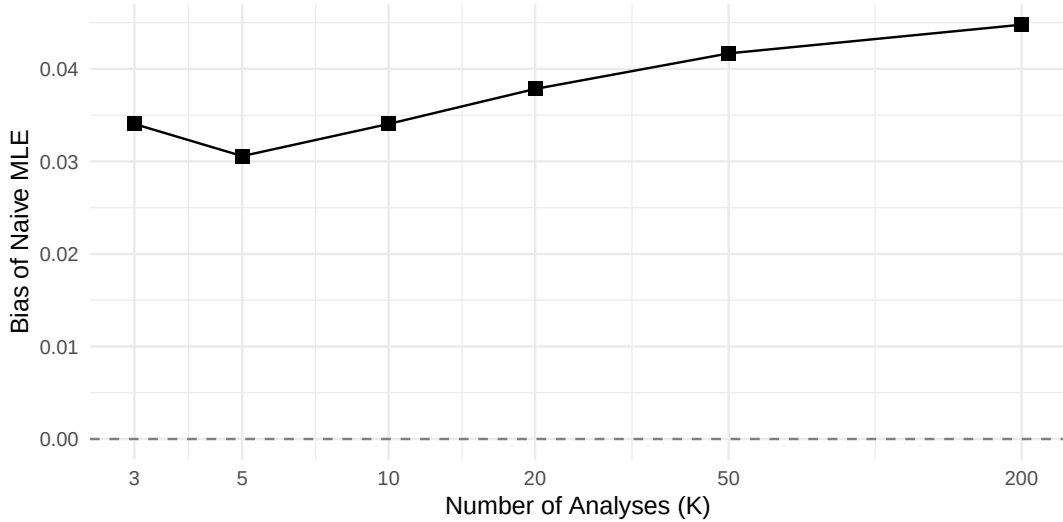


Figure 4: Naive MLE bias at the stopping time as a function of K. More frequent monitoring increases the conditional bias at stopping because the test statistic is more likely to cross the boundary at an extreme point. The bias remains small in absolute terms and tends to increase with K, within Monte Carlo noise (see MCSE(Bias) in the table above).

A well-known consequence of optional stopping is that the naive maximum likelihood estimate at the stopping time is biased away from the null²⁰. The figure above shows that this bias tends to increase with K (0.046 at $K = 50$ versus 0.044 at $K = 200$, a reversal within Monte Carlo noise: MCSE(Bias) is on the order of 0.003-0.004 at these K, so strict monotonicity is not established): more frequent monitoring creates more opportunities for the test statistic to cross the boundary at an extreme realization. However, two observations temper this concern. First, the absolute magnitude of the bias remains modest even at $K = 200$. Second, the bias is a known and correctable phenomenon. Several approaches to bias-corrected

estimation following sequential monitoring have been developed, including median unbiased estimation, conditional MLE methods, and bootstrap-based corrections. Practical adoption of high-frequency monitoring will require that these methods be packaged into accessible software rather than remaining the province of methodological specialists.

A.4 A Software Architecture for High-Frequency

Monitoring

The current generation of group sequential software – gsDesign²⁶, rpact²⁵, and the Lan-DeMets boundary programs – was designed for the $K = 3-5$ regime. While these tools can technically accommodate larger K , they were not optimized for it. A purpose-built system for high-frequency monitoring would require three components:

Boundary computation engine. The spending function must be evaluated at $K = 20-50$ information fractions, and the corresponding critical boundaries computed via numerical integration or recursive application of the multivariate normal distribution. For moderate K (up to 50), existing algorithms based on the work of²⁸ scale adequately. For K approaching N , direct computation becomes prohibitive and the continuous boundary approximation of¹² should be used instead. A practical implementation would pre-compute boundaries for a fine grid of information fractions and interpolate at the actual analysis times.

Bias-corrected estimation module. At each interim analysis, the system should report not only the naive test statistic and p-value but also:

- The median unbiased estimate of the treatment effect, computed by inverting the stage-wise ordering of the sample space.
- A repeated confidence interval that maintains simultaneous coverage across all analysis times.
- The conditional power given the data observed so far, to inform DSMB deliberations about futility.

These quantities are computationally more demanding than the naive estimates but are feasible for K up to 50 with modern hardware.

Automated alerting layer. The most consequential practical innovation would be an alerting system that decouples statistical monitoring from DSMB scheduling. In the current paradigm, the DSMB meets at pre-specified calendar times, and the interim analysis is conducted at that meeting. In a high-frequency monitoring paradigm, the statistical comparison would run continuously (or at each new data lock), and the system would generate alerts at three thresholds:

1. *Efficacy signal* – the test statistic crosses the upper spending function boundary. This triggers an expedited DSMB meeting to review the unblinded data and consider early termination.
2. *Futility signal* – the conditional power falls below a pre-specified threshold (e.g., 10–20%). This triggers a DSMB review of whether to continue enrollment.

529 3. *Safety signal* – a secondary monitoring rule for adverse events crosses a safety boundary.
530 This triggers immediate safety review regardless of the efficacy analysis schedule.

531 The DSMB retains full authority over all stopping decisions. The system merely ensures that
532 evidence is not accumulating unexamined between scheduled meetings. This is particularly
533 relevant for trials in which enrollment is fast relative to the DSMB meeting schedule: if 50
534 patients enroll between quarterly meetings, a group sequential design with $K = 5$ may miss
535 the opportunity to stop the trial at the statistically optimal time.

536 A.5 The Diminishing Marginal Cost of More Analyses

537 A counterargument to high-frequency monitoring is that each interim analysis incurs admin-
538 istrative cost: data locks, database reconciliation, statistical programming, and DSMB coordi-
539 nation. This is true in the current paradigm but largely an artifact of it. If the monitoring
540 system is automated – data are ingested continuously from electronic data capture, bound-
541 aries are pre-computed, and the test statistic is updated programmatically – then the marginal
542 cost of the 21st analysis is approximately zero. The cost structure shifts from per-analysis
543 overhead to a fixed upfront investment in infrastructure. For large trials where the cost per
544 enrolled patient is high and the potential savings from early termination are substantial, this
545 investment is easily justified.

546 A.6 Practical Recommendations

547 Based on the simulation results and the considerations above, we offer the following concrete
548 recommendations:

- 549 1. **Default to $K = 20$ rather than $K = 5$.** For trials with electronically captured end-
550 points and turnaround times of less than one month, the spending function framework
551 supports 20 planned analyses with no loss of type I error control and meaningful gains
552 in expected sample size. The practical overhead is minimal if the monitoring system is
553 automated.
- 554 2. **Use the O’Brien-Fleming spending function.** At high K , the O’Brien-Fleming
555 spending function spends negligible alpha at early analyses, so the cost of ‘looking more
556 often’ is almost entirely borne by a trivially small increase in the maximum sample size
557 (typically less than 1% inflation relative to the fixed-sample design).
- 558 3. **Report bias-corrected estimates at every analysis.** The naive MLE should never
559 be the sole estimate presented to a DSMB. At minimum, report the median unbiased
560 estimate and the repeated confidence interval alongside the naive values.
- 561 4. **Build the alerting system first.** The most impactful near-term investment is not in
562 novel statistical methodology but in software engineering: a system that ingests data,
563 updates the test statistic, evaluates it against pre-computed boundaries, and generates
564 structured alerts when thresholds are crossed. This system should produce a standard-
565 ized DSMB report at each triggered analysis, including the spending function status

(cumulative alpha spent), the bias-corrected estimate, conditional power, and a recommendation for whether a DSMB meeting is warranted.

5. **Treat K as a design parameter to be optimized.** Rather than choosing K by convention (the current default of 3–5), select K by explicitly trading off expected sample size savings against infrastructure cost. The diminishing-returns curve from the simulation provides the empirical basis for this trade-off.

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A remediation plan is recorded in the audit file and will be taken up in a subsequent revision.

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